

TIME SERIES ANALYSIS-Lecture-XVII

THIRD YEAR-SEMESTER –I I

2-CREDITS

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Seasonal Autoregressive Integrated Moving Average Models (SARIMA)

In practice many time series contain seasonal fluctuations. For example, with monthly data $s=12$ and quarterly data $s=4$.

SARIMA Models

$$SARIMA(p, d, q)(P, D, Q)_s$$

$$\phi_p(B) \Phi_Q(B^s) (1-B)^d (1-B^s)^D X_t = \theta_q(B) \Theta_Q(B^s) Z_t \rightarrow (1)$$

$$\phi_p(B) = 1 - \phi_1 B - \phi_2 B^2 - \phi_3 B^3 - \dots - \phi_p B^p$$

$$\Phi_P(B^s) = 1 - \Phi_1(B^s) - \phi_2(B^s)^2 - \Phi_3(B^s)^3 - \dots - \Phi_P(B^s)^P$$

$$\theta_q(B) = 1 + \theta_1 B + \theta_2 B^2 + \theta_3 B^3 - \dots + \theta_q B^q$$

$$\Theta_Q(B^s) = 1 + \Theta_1(B^s) + \Theta_2(B^s)^2 + \Theta_3(B^s)^3 - \dots + \Theta_Q(B^s)^Q$$

SARIMA Models

$$\phi_p(B) \Phi_P(B^s) \underbrace{(1-B)^d (1-B^s)^D}_{W_t} X_t = \theta_q(B) \Theta_Q(B^s) Z_t$$

$$W_t = (1-B)^d (1-B^s)^D X_t$$

$$\phi_p(B) \Phi_P(B^s) W_t = \theta_q(B) \Theta_Q(B^s) Z_t \quad ; \quad W_t = \nabla^d \nabla_s^D X_t$$

Example I SARIMA Models

Write the model for $SARIMA(1, 1, 1)(1, 1, 1)_4$

$$\phi_p(B) \Phi_Q(B^s) (1-B)^d (1-B^s)^D X_t = \theta_q(B) \Theta_Q(B^s) Z_t \rightarrow (1)$$

$$p = 1, q = 1, d = 1, P = 1, Q = 1, D = 1, s = 4$$

$$\phi_1(B) \Phi_1(B^4) (1-B)^1 (1-B^4)^1 X_t = \theta_1(B) \Theta_1(B^4) Z_t$$

Example I SARIMA Models

$$SARIMA(p, d, q)(P, D, Q)_s \Rightarrow SARIMA(1, 1, 1)(1, 1, 1)_4$$

$$\phi_p(B) = 1 - \phi_1 B - \phi_2 B^2 - \phi_3 B^3 - \dots - \phi_p B^p$$

$$\therefore \phi_1(B) = (1 - \phi_1 B)$$

$$\Phi_P(B^s) = 1 - \Phi_1(B^s) - \phi_2(B^s)^2 - \Phi_3(B^s)^3 - \dots - \Phi_P(B^s)^P$$

$$\therefore \Phi_1(B^4) = (1 - \Phi_1 B^4)$$

$$\theta_q(B) = 1 + \theta_1 B + \theta_2 B^2 + \theta_3 B^3 - \dots + \theta_q B^q$$

$$\therefore \theta_1(B) = (1 + \theta_1 B)$$

$$\Theta_Q(B^s) = 1 + \Theta_1(B^s) + \Theta_2(B^s)^2 + \Theta_3(B^s)^3 - \dots + \Theta_Q(B^s)^Q$$

$$\therefore \Theta_1(B^1) = (1 + \Theta_1 B^4)$$

Example I SARIMA Models

We know that SARIMA(1, 1, 1) (1, 1, 1)_{s=4}

has d=1, D=1 and s=4

$$\underbrace{(1 - \phi_1 B)}_{p=1} \underbrace{(1 - \Phi_1 B^4)}_{P=1} \underbrace{(1 - B)(1 - B^4)}_{W_t} X_t = \underbrace{(1 + \theta_1 B)}_{q=1} \underbrace{(1 + \Theta_1 B^4)}_{Q=1} Z_t$$

$$\therefore W_t = \nabla^1 \nabla_4^1 X_t = (1 - B)^1 (1 - B^4)^1 X_t = (1 - B)(1 - B^4) X_t$$

Example I SARIMA Models

SARIMA(1, 1, 1) (1, 1, 1)_s=4

$$(1 - \phi_1 B)(1 - \Phi_1 B^4)(1 - B)(1 - B^4)X_t = (1 + \theta_1 B)(1 + \Theta_1 B^4)Z_t$$

or

$$(1 - \phi_1 B)(1 - \Phi_1 B^4)W_t = (1 + \theta_1 B)(1 + \Theta_1 B^4)Z_t; \quad W_t = (1 - B)(1 - B^4)X_t$$

Example II SARIMA Models

Write the model for $SARIMA(2, 1, 1)(1, 1, 2)_{12}$

$$\phi_p(B) \Phi_Q(B^s) (1-B)^d (1-B^s)^D X_t = \theta_q(B) \Theta_Q(B^s) Z_t \rightarrow (1)$$

$$p = 2, q = 1, d = 1, P = 1, Q = 2, D = 1, s = 12$$

$$\phi_2(B) \theta_1(B) W_t = \Phi_1(B) \Theta_2(B) Z_t$$

Example II SARIMA Models

$$\phi_2(B) = 1 - \phi_1 B - \phi_2 B^2$$

$$\Phi_1(B) = 1 - \Phi_1(B^{12})$$

$$\theta_1(B) = 1 + \theta_1 B$$

$$\Theta_2(B) = 1 + \Theta_1(B^s) + \Theta_2(B^{12})^2$$

Example II SARIMA Models

$$W_t = \nabla^1 \nabla_{12}^1 X_t$$

$$W_t = (1 - B)(1 - B^{12})X_t$$

$$\underbrace{(1 - \phi_1 B - \phi_2 B^2)}_{p=2} \underbrace{(1 + \theta_1 B)}_{q=1} \underbrace{(1 - B)}_{d=1} \underbrace{(1 - B^{12})}_{D=1} X_t = \underbrace{(1 - \theta_1 B)}_{q=1} \underbrace{[(1 + \Theta_1 B^{12} + \Theta_2 (B^{12})^2)]}_{Q=2} Z_t$$

$$(1 - \phi_1 B - \phi_2 B^2)(1 + \theta_1 B)(1 - B)(1 - B^{12})X_t = (1 - \theta_1 B)[(1 + \Theta_1 B^{12} + \Theta_2 (B^{12})^2)]Z_t$$

or

$$(1 - \phi_1 B - \phi_2 B^2)(1 + \theta_1 B)W_t = (1 - \theta_1 B)[(1 + \Theta_1 B^{12} + \Theta_2 (B^{12})^2)]Z_t$$

$$W_t = (1 - B)(1 - B^{12})X_t$$

Seasonal Autoregressive Models

SAR(1)

$$SARIMA(0, 0, 0)(1, 0, 0)_{s=12}$$

$$p = 0, P = 1, q = 0, Q = 0, d = 0, D = 0,$$

$$\phi_p(B) \Phi_Q(B^s) (1-B)^d (1-B^s)^D X_t = \theta_q(B) \Theta_Q(B^s) Z_t$$

$$\Phi_1(B^s) X_t = Z_t$$

$$\Phi_1(B^s) = 1 - \Phi_1(B^s)$$

$$(1 - \Phi_1 B^s) X_t = Z_t$$

Seasonal Autoregressive Models

SAR(1)

Therefore, SAR(1) model for monthly data

$$(1 - \Phi_1 B^{12})X_t = Z_t$$

$$X_t = \Phi_1 B^{12} X_t + Z_t$$

$$X_t = \Phi_1 X_{t-12} + Z_t$$

SAR(1), has only one significant parameter.

Seasonal Moving Average Models

SMA(1)

$$SARIMA(p, d, q)(P, D, Q)_s$$

$$\phi_p(B) \Phi_Q(B^s) (1-B)^d (1-B^s)^D X_t = \theta_q(B) \Theta_Q(B^s) Z_t$$

$$SARIMA(0, 0, 0)(0, 0, 1)_{12}.$$

$$p = 0, P = 0, q = 0, Q = 1, d = 0, D = 0$$

Seasonal Moving Average Models

SMA(1)

$$\Theta_1(B^{12}) = 1 + \Theta_1 B^{12}$$

$$X_t = (1 + \Theta_1 B^{12})Z_t$$

$$X_t = Z_t + \Theta_1 B^{12} Z_t$$

$$X_t = Z_t + \Theta_1 Z_{t-12}$$

SMA(1), has only one significant parameter.

SAR(1)

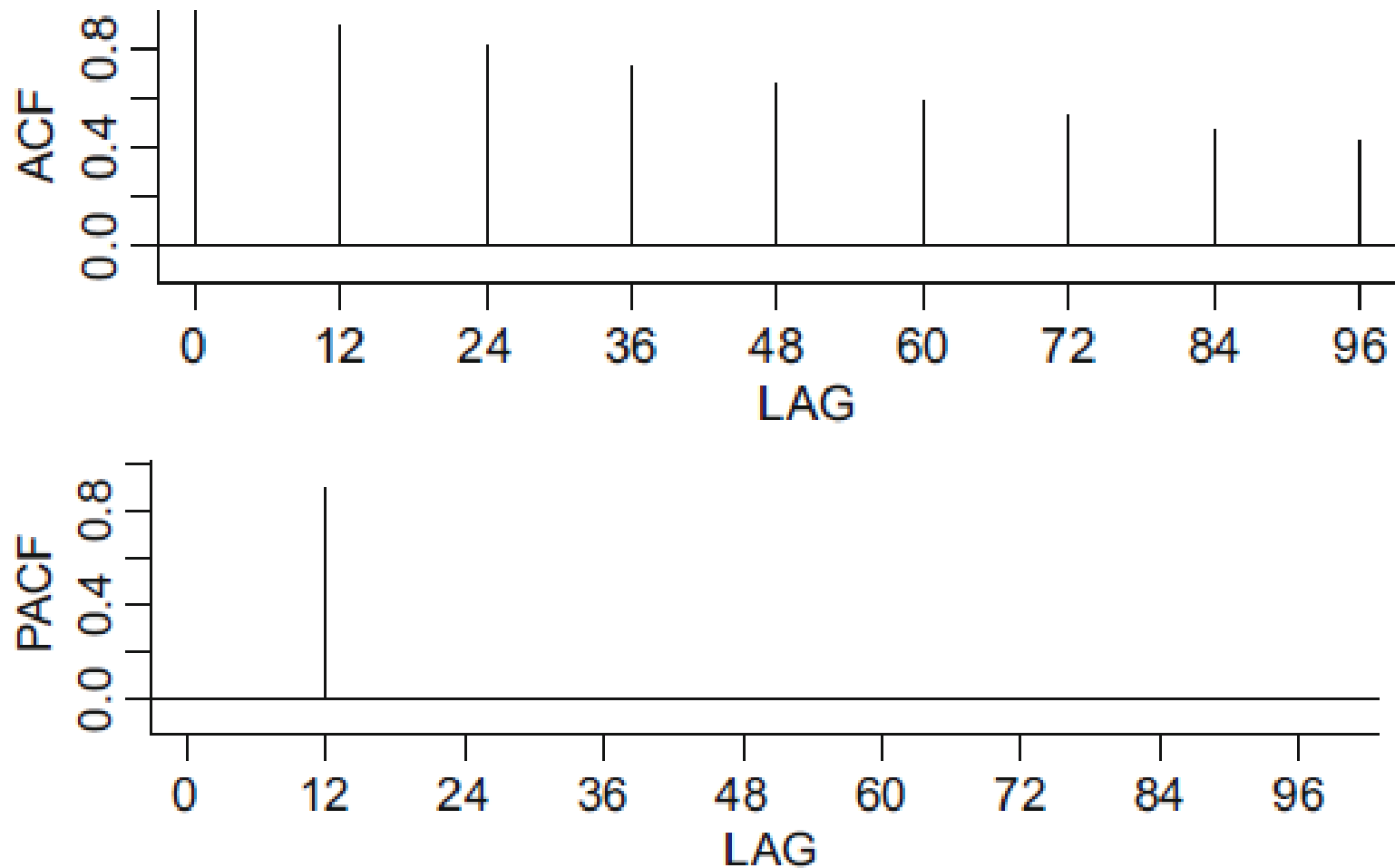


Fig. 1: ACF and PACF for SAR (1) (Shumway & Stoffer, 2017)

SMA(1)

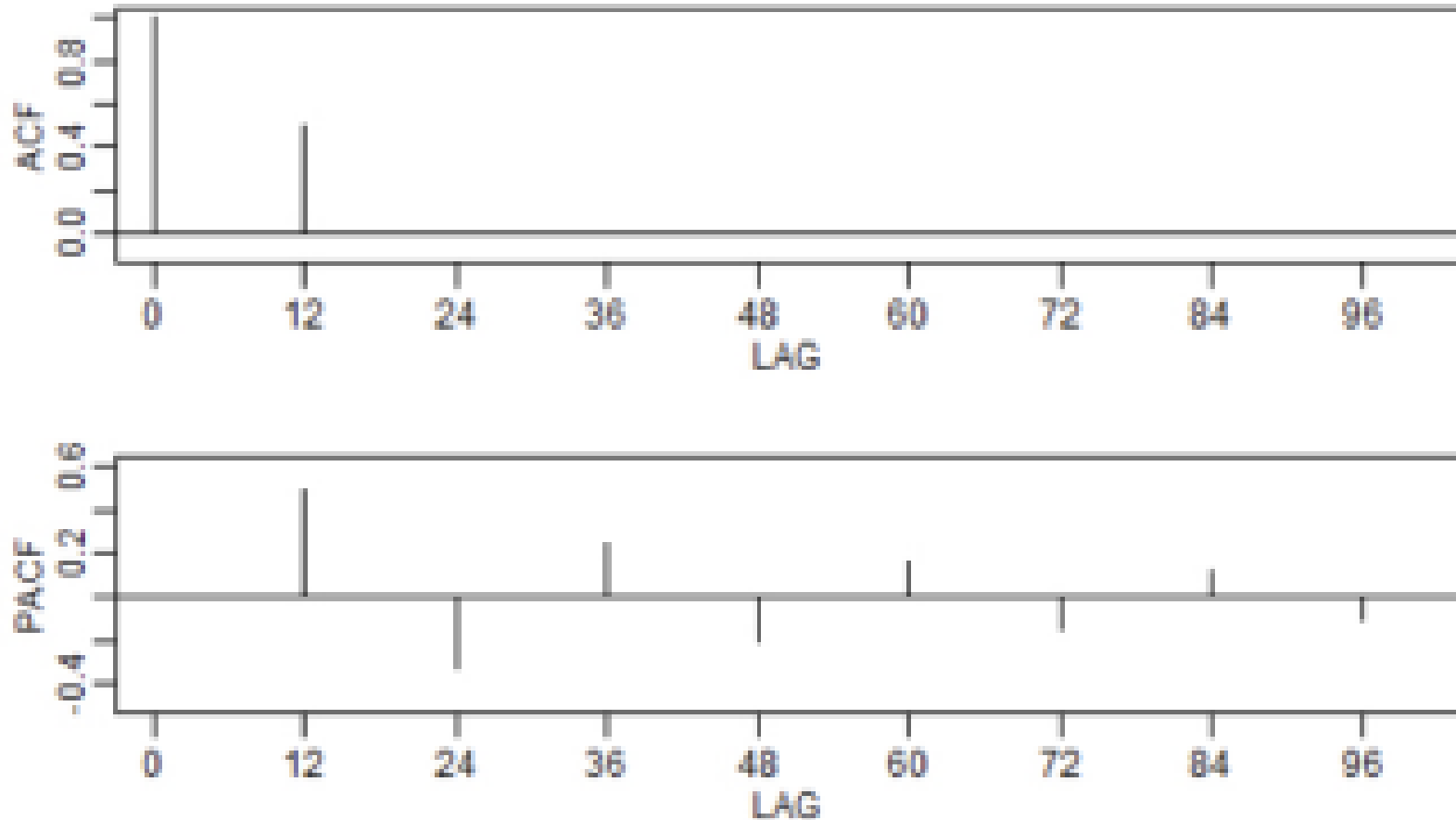


Fig. 2: ACF and PACF for SMA (1) (Notes: Stoffer, D.)